

The Brunn-Minkowski Inequality

following R. J. Gardner

The most elementary Brunn-Minkowski inequality is that on $\mathbb{R}$ with the default Lebesgue measure.

THM 1. BRUNN-MINKOWSKI INEQUALITY ON $\mathbb{R}$.

Let $\lambda \in (0, 1)$ and $X, Y \subseteq \mathbb{R}$ be nonempty bounded measurable sets s.t. $(1 - \lambda)X + \lambda Y$ is also measurable. Then,

$$\text{vol}[(1 - \lambda)X + \lambda Y] \geq (1 - \lambda) \text{vol } X + \lambda \text{vol } Y$$

Proof.

W.l.o.g. let X, Y be compact and the general noncompact case follows by using a compact exhaustion. By taking the union of covers $X + Y$ is also compact. As measures of sets are translation-invariant translate $X \subseteq \mathbb{R}^-$ and $Y \subseteq \mathbb{R}^+$ so $X + Y \supseteq X \sqcup Y$. Dilations preserve this relation by multiplying elements on both sides $(1 - \lambda)X + \lambda Y \supseteq (1 - \lambda)X \sqcup \lambda Y$. Hence,

$$\text{vol}[(1 - \lambda)X + \lambda Y] \geq \text{vol}[(1 - \lambda)X \sqcup \lambda Y] = (1 - \lambda) \text{vol } X + \lambda \text{vol } Y$$

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There are several variants and generalizations of this classic theorem.

We preliminarily define several objects that we will use extensively.

DEF 1. Let $\mathbb{B}^n$ denote the unit n -ball of $\mathbb{R}^n$ and $\mathbb{S}^{n-1} = \partial \mathbb{B}^n$ the unit $(n - 1)$ -sphere. For $\mathbf{u} \in \mathbb{S}^{n-1}$ denote $\mathbf{u}^\perp$ to be the hyperplane tangent to $\mathbb{S}^{n-1}$ and normal to $\mathbf{u}$. We identify the k -dimensional Lebesgue measure vol_k with the k -dimensional Hausdorff measure, and so integration w.r.t. $d\mathbf{x}$ will denote integration w.r.t vol_k (for the appropriate k) and $d\mathbf{u}$ will denote integration w.r.t. vol_{n-1} .

DEF 2. For $X, Y \subseteq \mathbb{R}^n$ their Minkowski sum is defined to be,

$$X + Y = \{x + y \mid x \in X, y \in Y\}$$

Similarly for $\lambda \in \mathbb{R}$ their ‘dilation’ or ‘reflection then dilation’ is defined,

$$\lambda X = \{\lambda x \mid x \in X\}$$

And so when $\lambda \in (0, 1)$ the set $(1 - \lambda)X + \lambda Y$ is called the ‘convex combination of X and Y ’.

DEF 3. Define the surface area of a set $X \subseteq \mathbb{R}^n$ to be,

$$\text{sur } X = \lim_{\epsilon \rightarrow 0} \frac{\text{vol}[X + \epsilon \mathbb{B}^n] - \text{vol } X}{\epsilon}$$

We use this definition when X is a convex body or is compact with piecewise C^1 boundary.

DEF 4. A function f is concave on a convex set X if,

$$f[(1 - \lambda)\mathbf{x} + \lambda\mathbf{y}] \geq (1 - \lambda)f[\mathbf{x}] + \lambda f[\mathbf{y}]$$

For all $\mathbf{x}, \mathbf{y} \in X$. This is equivalent to that $-f$ is convex. A nonnegative function is ‘log-concave’ if its logarithm is concave, or by exponentiating,

$$f[(1 - \lambda)\mathbf{x} + \lambda\mathbf{y}] \geq f[\mathbf{x}]^{1-\lambda} f[\mathbf{y}]^\lambda$$

And by the AM-GM inequality if a function is concave then it is log-concave.

DEF 5. Define the level-set of a nonnegative measurable function and a value,

$$\{f \geq t\} = \{\mathbf{x} \in \mathbb{R}^n \mid f[\mathbf{x}] \geq t\}$$

We have by Fubini’s Theorem,

$$\int_{\mathbb{R}^n} f \, d\mathbf{x} = \int_{\mathbb{R}^n} \int_0^{f[\mathbf{x}]} dt \, d\mathbf{x} = \int_0^\infty \int_{\{f \geq t\}} d\mathbf{x} \, dt = \int_0^\infty \text{vol}\{f \geq t\} \, dt$$

The intermediate equality,

$$f[\mathbf{x}] = \int_0^\infty 1_{\{f \geq t\}}[\mathbf{x}] \, dt$$

Is known as the layer-cake representation of f .

The next result is an analytic inequality instrumental to proving Brunn-Minkowski,

THM 2. PREKOPA-LEINDLER INEQUALITY ON $\mathbb{R}$

Let $\lambda \in (0, 1)$ and f, g, h be nonnegative integrable functions on $\mathbb{R}$ satisfying,

$$h[(1 - \lambda)x + \lambda y] \geq f[x]^{1-\lambda} g[y]^\lambda$$

For all $x, y \in \mathbb{R}$. Then,

$$\int_{\mathbb{R}} h \geq \left(\int_{\mathbb{R}} f \right)^{1-\lambda} \left(\int_{\mathbb{R}} g \right)^\lambda$$

Proof.

W.l.o.g. we normalize f and g by their supremum so that,

$$\sup_{\mathbb{R}} f = \sup_{\mathbb{R}} g = 1$$

Note for $t \in \mathbb{R}^+$ that $f[x], g[y] \geq t$ jointly means $h[(1-\lambda)x + \lambda y] \geq t$ by the assumption for all λ . Written in terms of level sets,

$$\{h \geq t\} \supseteq (1-\lambda)\{f \geq t\} + \lambda\{g \geq t\}$$

Therefore,

$$\begin{aligned} \int_{\mathbb{R}} h &\geq \int_0^1 \text{vol}\{h \geq t\} dt \\ &\geq \int_0^1 \text{vol}[(1-\lambda)\{f \geq t\} + \lambda\{g \geq t\}] dt \\ &\geq \int_0^1 (1-\lambda) \text{vol}\{f \geq t\} + \lambda \text{vol}\{g \geq t\} dt \\ &= (1-\lambda) \int_0^1 f + \lambda \int_0^1 g \\ &= \left(\int_0^1 f\right)^{1-\lambda} \left(\int_0^1 g\right)^{\lambda} \end{aligned}$$

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The n -dimensional case uses the one-dimensional case in induction.

THM 3. PREKOPA-LEINDLER INEQUALITY ON $\mathbb{R}^n$

Let f, g, h be nonnegative measurable functions on $\mathbb{R}^n$ satisfying,

$$h[(1-\lambda)\mathbf{x} + \lambda\mathbf{y}] \geq f[\mathbf{x}]^{1-\lambda} g[\mathbf{y}]^{\lambda}$$

For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then,

$$\int_{\mathbb{R}^n} h \geq \left(\int_{\mathbb{R}^n} f\right)^{1-\lambda} \left(\int_{\mathbb{R}^n} g\right)^{\lambda}$$

Proof.

The proof proceeds by induction on n . The base case is given by THM 2. Suppose true for all less than n . For all $s \in \mathbb{R}$ define respectively,

$$f_s, g_s, h_s : \mathbb{R}^{n-1} \rightarrow \mathbb{R}, \quad \mathbf{z} \mapsto f[\mathbf{z}, s], g[\mathbf{z}, s], h[\mathbf{z}, s]$$

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{n-1}$ and $a, b \in \mathbb{R}$ these functions satisfy our assumption in,

$$\begin{aligned} h_{(1-\lambda)a + \lambda b}[(1-\lambda)\mathbf{x} + \lambda\mathbf{y}] &= h[(1-\lambda)\mathbf{x} + \lambda\mathbf{y}, (1-\lambda)a + \lambda b] \\ &= h[(1-\lambda)(\mathbf{x}, a) + \lambda(\mathbf{y}, b)] \\ &\geq f[\mathbf{x}, a]^{1-\lambda} g[\mathbf{y}, b]^{\lambda} \\ &= f_a[\mathbf{x}]^{1-\lambda} g_b[\mathbf{y}]^{\lambda} \end{aligned}$$

So we are able to use our inductive hypothesis,

$$\int_{\mathbb{R}^{n-1}} h_{(1-\lambda)a+\lambda b} \geq \left(\int_{\mathbb{R}^{n-1}} f_a \right)^{1-\lambda} \left(\int_{\mathbb{R}^{n-1}} g_b \right)^\lambda$$

If we define the quantities,

$$F[s] = \int_{\mathbb{R}^{n-1}} f_s, \quad G[s] = \int_{\mathbb{R}^{n-1}} g_s, \quad H[s] = \int_{\mathbb{R}^{n-1}} h_s$$

Then the inequality above gives $H[(1-\lambda)a+\lambda b] \geq (1-\lambda)F[a] + \lambda G[b]$ our antecedent on $\mathbb{R}$. Hence using the inductive hypothesis,

$$\int_{\mathbb{R}^n} h = \int_{\mathbb{R}} H \geq \left(\int_{\mathbb{R}} F \right)^{1-\lambda} \left(\int_{\mathbb{R}} G \right)^\lambda = \left(\int_{\mathbb{R}^n} f \right)^{1-\lambda} \left(\int_{\mathbb{R}^n} g \right)^\lambda$$

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Let $f_i \in L^{p_i}[\mathbb{R}^n]$ nonnegative functions where $p_i \geq 1$ and $\sum_i p_i^{-1} = 1$. We have the classic Holder's Inequality,

$$\int_{\mathbb{R}^n} \prod_{i=1}^m f_i \leq \prod_{i=1}^m \|f_i\|_{p_i} = \prod_{i=1}^m \left(\int_{\mathbb{R}^n} f_i^{p_i} \right)^{1/p_i}$$

For $\lambda \in (0, 1)$, in the particular case $m = 2$, $1/p_1 = 1 - \lambda$, $1/p_2 = \lambda$ applied to $f^{1-\lambda}$, g^λ we have,

$$\int_{\mathbb{R}^n} f^{1-\lambda} g^\lambda \leq \left(\int_{\mathbb{R}^n} f \right)^{1-\lambda} \left(\int_{\mathbb{R}^n} g \right)^\lambda$$

By using the supremum of the l.h.s. the Prekopa-Leindler Inequality may also be written,

$$\int_{\mathbb{R}^n} \sup \{ f[\mathbf{x}]^{1-\lambda} g[\mathbf{y}]^\lambda \mid (1-\lambda)\mathbf{x} + \lambda\mathbf{y} = \mathbf{z} \} d\mathbf{z} \geq \left(\int_{\mathbb{R}^n} f \right)^{1-\lambda} \left(\int_{\mathbb{R}^n} g \right)^\lambda$$

Or, inductively,

$$\int_{\mathbb{R}^n} \sup \left\{ \prod_{i=1}^m f_i \mid \sum_{i=1}^m \frac{x_i}{p_i} = z_i \right\} d\mathbf{z} \geq \prod_{i=1}^m \|f_i\|_{p_i} = \prod_{i=1}^m \left(\int_{\mathbb{R}^n} f_i^{p_i} \right)^{1/p_i}$$

So the Prekopa-Leindler Inequality is in some sense the converse to Holder's Inequality.